FDS EXPERIMENT-16

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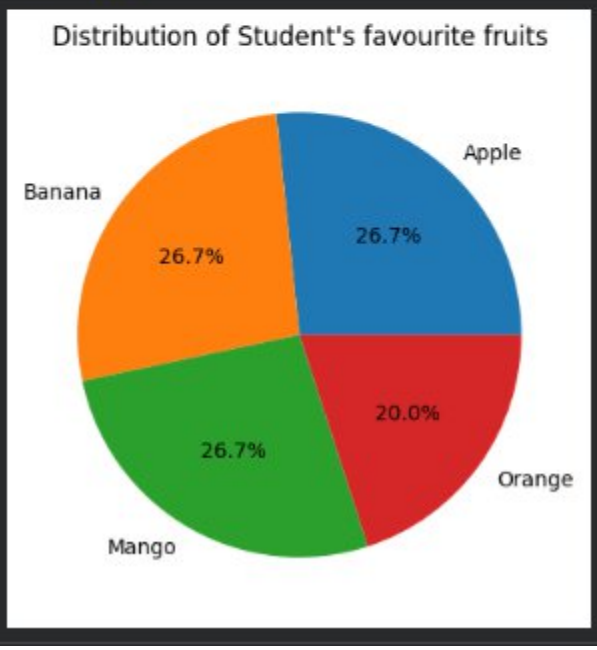
Experiment to understand EDA Quantitative and Qualitative Analysis.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from collections import Counter
```

```
#Nominal Data
fruits = [
    "Apple","Banana","Mango","Apple","Orange",
    "Mango","Banana","Apple","Mango","Orange",
    "Banana","Mango","Apple","Orange","Banana"
]

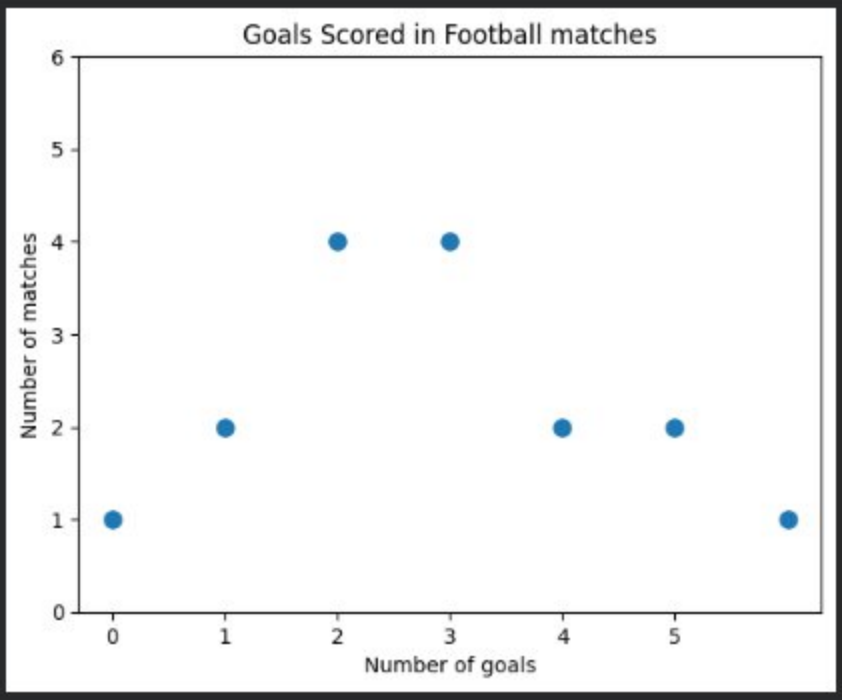

fruits_counts = Counter(fruits)
```

```
plt.pie(fruits_counts.values(), labels = fruits_counts.keys(),autopct = '%1.1f%%')
plt.title("Distribution of Student's favourite fruits")
plt.show()
```



```python
#Discrete data
goal_count = [2,5,3,1,4,2,3,0,3,2,5,1,4,3,3,2]
frequency = Counter(goal_count)
values = sorted(frequency.keys())
frequencies = [frequency[value] for value in values]
```

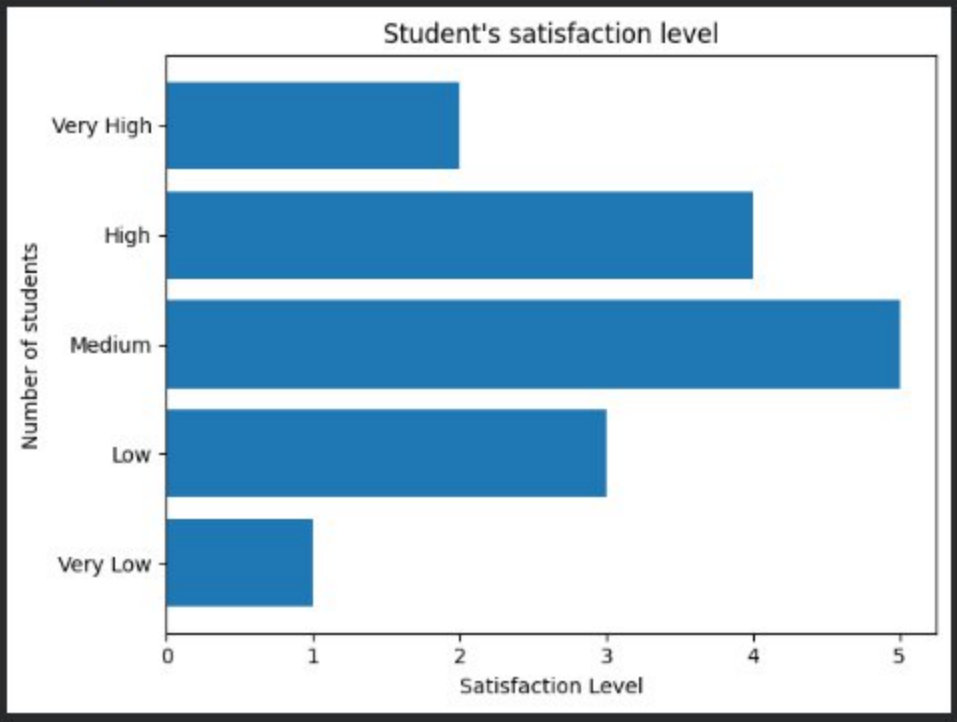
```python
plt.plot(values,frequencies,marker = 'o',linestyle = 'None',markersize = 8)
plt.title("Goals Scored in Football matches")
plt.xlabel("Number of goals")
plt.ylabel("Number of matches")
plt.xticks(range(0,6))
plt.yticks(range(0,7))
plt.show()
```



```python
#Ordinal Data
satisfaction = [
    "Low","Medium","High","Medium","Very High",
    "Low","High","Medium","Very High","High",
    "Low","Medium","High","Very Low","Medium"
]

category_order = ["Very Low","Low","Medium","High","Very High"]
count = Counter(satisfaction)
```

```python
frequencies = [count[category] for category in category_order]
plt.barh(category_order,frequencies)
plt.title("Student's satisfaction level")
plt.xlabel("Satisfaction Level")
plt.ylabel("Number of students")
plt.tight_layout()
plt.show()
```



```
#Continuous data

heights = [
    150.5,162.3,155.8,170.2,165.7,
    158.4,172.6,160.1,168.9,154.6,
    163.5,175.2,159.7,166.4,152.8
]
```

```
plt.hist(heights,bins = 5,edgecolor = 'black')
plt.title("Distribution of Student's Heights")
plt.xlabel("Height (cm)")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()
```

